

Python For Finance Professionals

Portfolio Optimization

Objectives

By the end of this session, participants will be able to:

Construct portfolios with 2 securities and extend to multiple securities.

Calculate expected return and portfolio risk using variance, covariance, and correlation.

Evaluate performance using key metrics: **Sharpe Ratio, Alpha, and Beta**.

Explain and apply the **Capital Asset Pricing Model (CAPM)** to assess expected returns.

Measure systematic vs. unsystematic risk and understand diversification benefits.

Visualize the Efficient Frontier and identify optimal portfolios using Python.

The CAPM

Capital Asset Pricing Model

Portfolio Overview

A basket of assets

May hold stocks, bonds, cash, commodities, derivatives, etfs

Investors aim for a return by mixing these securities in a manner that reflects

- Their appetite for risk

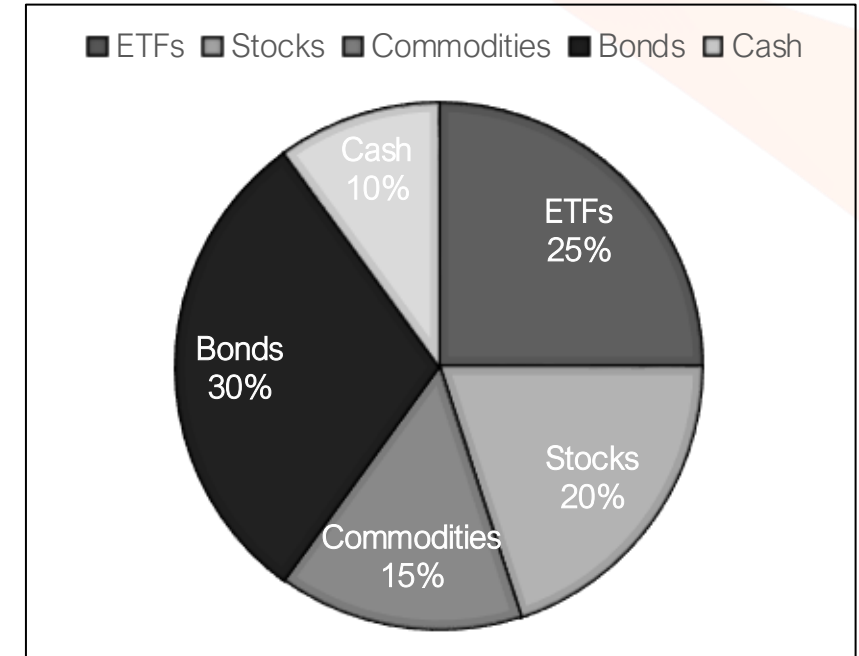
- Their financial goals

- Their time horizon

- ...

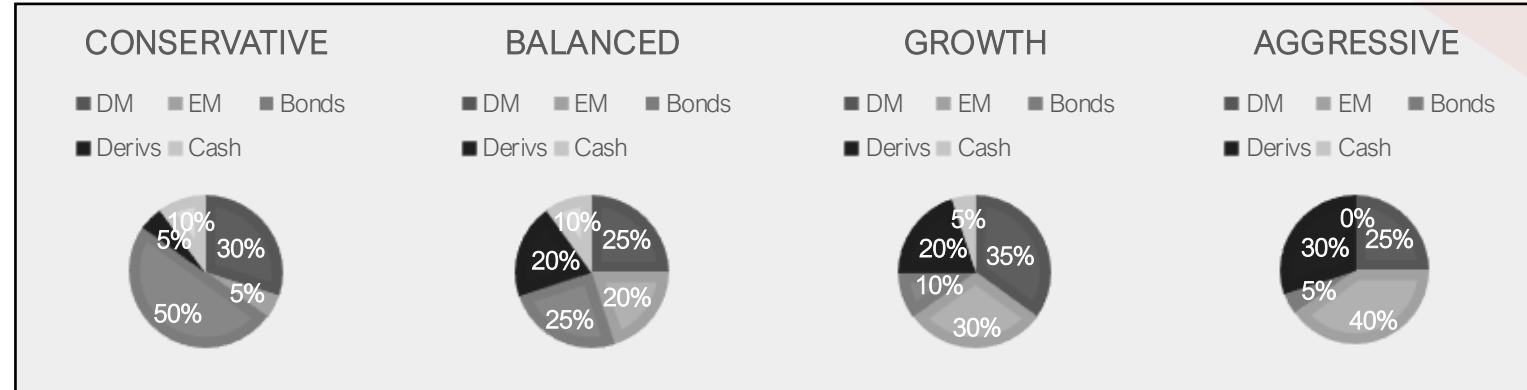
A key concept in portfolio management is diversification

Don't carry all your eggs in one basket



Portfolio Overview

Developed Markets
Emerging Markets
Derivatives
Cash



Annual Return %

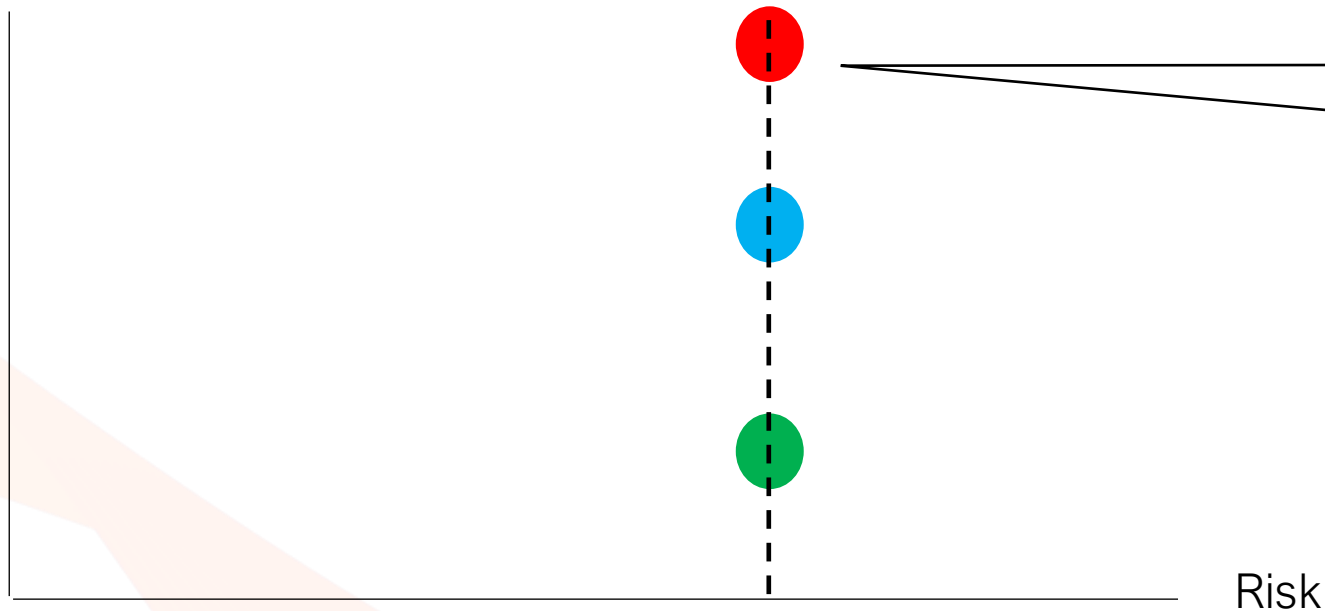
	CONSERVATIVE	BALANCED	GROWTH	AGGRESSIVE
Average	6.01	7.98	8.97	9.64
Best 12 months	31.06	76.57	109.55	136.07
Worst 12 months	-17.67	-40.64	-52.92	-60.78
Best 5 years	17.23	23.14	27.27	31.91
Worst 5 years	0.37	-6.18	-10.43	-13.78

Portfolio Overview

Assume investors are rational

Irrespective of their motivation, they want the **maximum** amount of return for the **least** amount of risk

Return



Irrespective of their appetite for Risk
A rational investor would choose this
return
i.e. the maximum return for a given
amount of risk

Portfolio Overview

Assume investors are rational

Irrespective of their motivation, they want the **maximum** amount of return for the **least** amount of risk

Return



Irrespective of their appetite for Risk
A rational investor would choose this return
i.e. the maximum return for the minimum amount
of risk

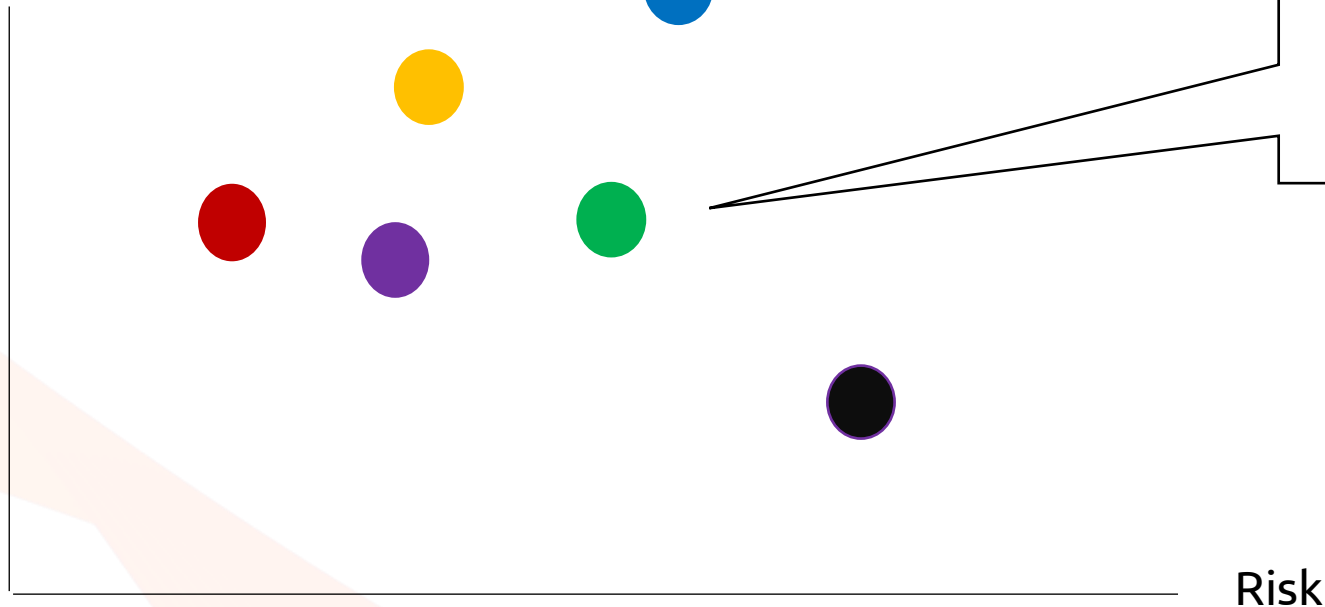
Risk

Portfolio Overview

Assume investors are rational

Irrespective of their motivation, they want the **maximum** amount of return for the **least** amount of risk

Return



What about this set of returns ?

Risk

Problem

Given you have **\$100,000** to invest for 1 year.

Treasury bills yield 5% (TBills pay a lot less, but this is for demonstration purposes only)

One alternative is to invest in TBills for no risk where the expected return is 5%.

Another alternative is to invest in stock.

Suppose the possible outcomes from investing in stock are

Probability	Return (%)
.05	50
.25	30
.4	10
.25	-10
.05	-30

expected return is calculated by

$$(0.05 * 0.5) + (0.25 * 0.3) + (0.4 * 0.1) + (0.25 * -0.1) + (0.05 * -0.3) \\ = 0.1$$

May get \$50K return

May also lose \$30K

Quantifying Risk

Often std deviation is used

$$\sigma = \sqrt{E(R^2) - [E(R)]^2}$$

Expected Return - E(R)

$$\begin{aligned} &= (0.05 \times 0.5) + (0.25 \times 0.3) + (0.4 \times 0.1) + (0.25 \times -0.1) + (0.05 \times -0.3) \\ &= 0.1 \end{aligned}$$

$$[E(R)]^2 = 0.01$$

E(R²)

$$\begin{aligned} &= (0.05 \times 0.5^2) + (0.25 \times 0.3^2) + (0.4 \times 0.1^2) + (0.25 \times -0.1^2) + (0.05 \times -0.3^2) \\ &= 0.046 \end{aligned}$$

std deviation = 0.1897

Quantifying Risk

A portfolio consisting of 2 securities with returns R_1 and R_2

Put a proportion of money into the first ω_1

The remainder into the second $\omega_2 = 1 - \omega_1$

The return on the investment is $\omega_1 R_1 + \omega_2 R_2$

The portfolio expected return $\mu_p = \omega_1 \mu_1 + \omega_2 \mu_2$

where

μ_1 is the expected return on the first investment

μ_2 is the expected return on the second investment

The std deviation of such a portfolio is

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2\rho w_1 w_2 \sigma_1 \sigma_2}$$

Portfolio with 2 securities

Given the following

$$\begin{aligned}\mu_1 &= 10\% \quad \sigma_1 = 16\% \\ \mu_2 &= 15\% \quad \sigma_2 = 24\% \\ \rho &= 20\%\end{aligned}$$

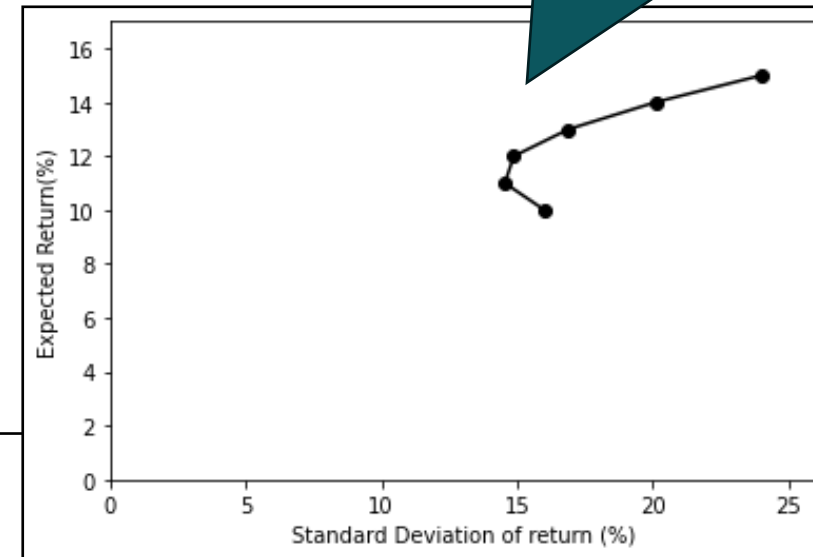
Most investors are risk averse

They want increased expected returns while reducing std deviation of return. They will want to move as far as possible in a **North West** direction. Analysis such as this helps investors make more informed investing decisions.

The expected return and standard deviation for this portfolio for weights of

(0%, 100%),
(20%, 80%),
(40%, 60%),
(60%, 40%),
(80%, 20%),
(100%, 0%)

ω_1	ω_2	μ_p	σ_p
0.0	1.0	15%	24.00%
0.2	0.8	14%	20.09%
0.4	0.6	13%	16.89%
0.6	0.4	12%	14.87%
0.8	0.2	11%	14.54%
1.0	0.0	10%	16.00%



Efficient Frontier

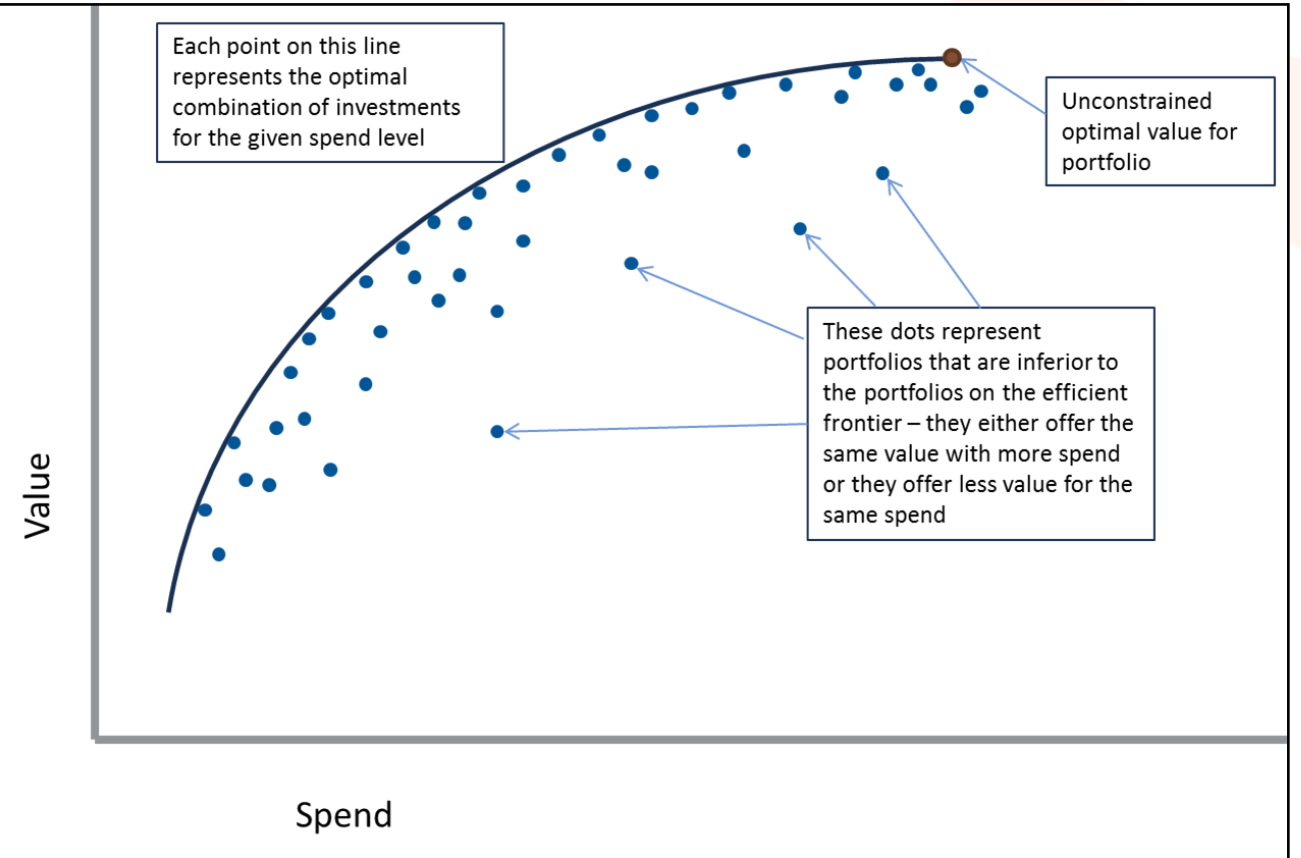
An investor can

- add a third investment to the portfolio
- combine it with any combination of the first 2 produce a new risk return.

They can repeat this with a 4th, 5th investment etc.

Eventually we reach the limit of how far North West we can go.

This is known as the **efficient frontier**



Beta

Beta is a measure of the volatility, systematic risk of a security or portfolio compared to the market as a whole.

It can only provide an investor with an approximation of how much risk the stock will add to a portfolio.

For beta to be meaningful, the stock should be related to the benchmark that is used in the calculation

Types of Beta

Equal to 1 – Price strongly correlated to market. Not likely to have excess returns

Less than 1 – Asset less volatile than market. Usually moves more slowly than market averages

Greater than 1 – More volatile than market – Will increase risk but also may increase expected return

Negative Beta – Inversely correlated to market

CAPM

In general

$$E(R) = R_F + \beta[E(R_M) - R_F]$$

This is also known as the capital asset pricing model

The excess expected return over risk-free rate required on investment is β times the excess expected return on market

For example:

If the risk free rate is 5% and the return on market portfolio is 10%

An investment of β 0 should have an expected return of 5%

An investment of β 0.5 should have an expected return of $0.05 + 0.5 \times (0.1 - 0.05) = 0.075$ (7.5%)

An investment of β 1.2 should have an expected return of $0.05 + 1.2 \times (0.1 - 0.05) = 0.11$ (11%)

Alpha

Assume that the portfolio manager produces a superior return for the amount of risk taken.

The extra return can be calculated using $R_P - R_F - \beta(R_M - R_F)$

If this is **positive**, it means the investment outperformed the market.

If its **negative**, it means the investment underperformed the market.

This is known as alpha α

Example

A portfolio manager has a portfolio with $\beta = 0.8$.

The 1 year risk free rate of return is 5%.

The return on the market for the year is 7%.

The portfolio manager's return is 9%.

The manager's α is $0.09 - 0.05 - 0.8 \times (0.07 - 0.05) = 0.024$

